

7	(a)	(i)	As the frame is in motion, the wire ST cuts the magnetic field and there is a rate of change of magnetic flux linkage The e.m.f. induced is proportional to the rate of change of flux linkage Hence there is an induced e.m.f. in the wire ST.	
		(ii)	As the frame oscillates, its <u>velocity will vary in a sinusoidal</u> manner. Therefore, the <u>rate of change of magnetic flux linkage will also change sinusoidally</u>. Since the emf across ST is proportional to the rate of change of magnetic flux linkage, it will also vary sinusoidally.	
	(b)	(i)	1 mark: sinusoidal e.m.f with <u>decreasing amplitude</u> The <u>induced current</u> results in a <u>magnetic force on the metal wire ST</u> which opposes the motion of the frame ASTB and hence the oscillation is <u>damped</u>. This affects the maximum speed and hence induced e.m.f in the frame. Therefore, amplitude of e.m.f decreases.	

		(ii)	<i>Case 1 (considering the whole loop)</i> The oscillation will be undamped and the amplitude will be unchanged because there is no e.m.f. induced in the frame and there will be no current in the loop and there is no electromagnetic damping. OR There is no change in the magnetic flux linkage, thus there is no emf induced. Thus, the graph is a straight horizontal line, cutting the emf = 0. <i>Case 2 (considering ST)</i> Net emf induced is zero, therefore no current within rod ST, thus, no damping force. The graph drawn in Fig 7.2 remains the same.	

8	(a)	(i)	Photoelectric effect is a phenomenon in which electrons are emitted from the surface of	
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